ADL4CV Project Report: Instant Surface Reconstruction

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Abstract

Recent advancements in 3D scene representation, particularly 2D Gaussian Splatting (2DGS), have demonstrated high-quality novel view synthesis and surface reconstruction. However, 2DGS relies on a sparse Structure-from-Motion (SfM) initialization followed by a reactive and slow densification process. In this work, we propose to accelerate high-quality mesh extraction by introducing a dense initialization strategy inspired by EDGS. Our method precomputes a dense set of 2D Gaussians using triangulated correspondences and enhances this initialization with estimated surface normals and depth filtering. Evaluated on the DTU dataset, our approach achieves state-of-the-art geometric accuracy in significantly less training time compared to vanilla 2DGS and other explicit methods.

1. Introduction

Reconstructing accurate 3D surfaces from multi-view images is a fundamental challenge in computer vision. While radiance fields like NeRF [5] and 3D Gaussian Splatting (3DGS) [2] offer photorealistic novel view synthesis, they often lack the structural accuracy required for downstream applications such as physics simulation and asset creation. 2D Gaussian Splatting (2DGS) [1] addresses this by utilizing oriented planar disks to better define surface geometry. However, 2DGS inherits the reliance on sparse Structure-from-Motion (SfM) initialization from 3DGS, necessitating a slow, reactive densification process to discover scene topology.

Recently, Eliminating Densification for 3DGS (EDGS) [3] proposed a shift by replacing the heuristic densification process with a dense initialization strategy, demonstrating that dense initialization can eliminate densification for 3DGS. However it remains centered on 3D Gaussians, thus lacking the surface-aware constraints required for high-fidelity meshing.

In this work, we propose to bridge the gap between the geometric rigor of 2DGS and the efficiency of EDGS. We introduce a framework for Fast Surface Reconstruction that adapts the dense initialization of EDGS to the 2DGS primitive. To make this adaptation successful, we introduce a prior-aware initialization. Instead of initializing disks with random orientations, we leverage normal estimation to align 2D Gaussians with the underlying surface geometry at iteration zero. Furthermore, we employ depth-based filtering to remove outliers from the dense matching process. Our contributions are as follows:

- We adapt the dense, triangulation-based initialization of EDGS to the 2DGS framework, eliminating the need for slow, iterative densification.
- We introduce a prior-aware initialization scheme that utilizes estimated surface normals and depth-consistent filtering to pre-align 2D Gaussians, significantly reducing the optimization search space.
- We demonstrate that our method achieves state-of-the-art mesh quality on the DTU dataset in under 2 minutes of training, a 5-10x speedup over vanilla 2DGS.

2. Related Work

2.1. Novel View Synthesis

Recent work has made significant advancements in Novel View Synthesis (NVS). NVS aims to reconstruct a 3D scene from 2D images and create renderings from unseen views. NeRF [5] utilizes a multi-layer perceptron (MLP) to encode the scene as a continuous volumetric function of density and color. While NeRF achieved unprecedented visual fidelity, its reliance on computationally intensive ray marching led to slow training and inference times. More recently, 3D Gaussian Splatting (3DGS)[2] moved from NeRF's implicit volumetric representation to an explicit point-based approach. By representing scenes as a collection of differentiable 3D Gaussians, this method achieves significantly faster optimization and real-time rendering without sacrificing visual fidelity.

2.2. Surface Reconstruction

While 3D Gaussian Splatting produces photorealistic view renderings, its 3D ellipsoids often lack consistency across different viewing angles. 2D Gaussian Splatting[1] aimed to tackle this issue by replacing 3D volumes with 2D disks. The 2D primitives force the geometry onto a thin manifold, providing the clear structural normals required for accurate

surface reconstruction while maintaining the speed advantages.

2.3. Improving Gaussian Initialization

The efficiency of Gaussian-based reconstruction depends heavily on the quality of the initial point cloud. Several recent works focus on accelerating the reconstruction process by moving away from traditional Structure-from-Motion (SfM) pipelines. QuickSplat[4] utilizes a feed-forward network to predict Gaussian parameters directly from images for near-instantaneous initialization. Similarly, EDGS (Eliminating Densification for Gaussian Splatting)[3] removes the need for iterative densification heuristics by establishing a dense initialization from triangulated correspondences.

3. Method

Our method aims to enhance the efficiency of 2D Gaussian Splatting based surface reconstruction by incorporating a dense initialization strategy. Specifically, we integrate the method introduced by EDGS[3] to generate a very dense and accurate point cloud. We then use the calculated points to initialize the parameters of the initial 2D Gaussians. After that, we improve the Gaussian model's accuracy using the standard iterative 2DGS optimization pipeline. We illustrate our method in Fig 1.

3.1. Surface Representation

We adopt 2D Gaussian Splatting (2DGS)[1] to represent the scene as a collection of oriented 2D Gaussian disks. This formulation is particularly suited for surface reconstruction, as it constrains geometry to a thin manifold and provides well-defined surface normals. Reconstructed images are synthesized through depth-sorted alpha-blending of these primitives' color and opacity. This representation ultimately allows for high-fidelity mesh extraction through TSDF fusion of the rendered depth maps. Our method proposes a dense initialization strategy that starts the optimization with a much more accurate set of 2D Gaussians, enabling faster convergence compared to the sparse SfM initialization used by 2DGS.

3.2. Dense Initialization

Point cloud generation For the initial point cloud generation, we use the method proposed by EDGS[3]. EDGS uses a pretrained dense matching network to establish pixel correspondences across views. The positions of the generated points are then calculated by triangulating between the camera poses of the respective views and the matched pixel coordinates.

After generating the initial points from matched pixel correspondences, we filter outlier points with per-view estimated depth maps. Then we initialize the parameters for the 2D Gaussians for the remaining 3D points. To refine the initial reconstruction accuracy, we estimate normals and use a heuristic based on local point density to set the initial scale and opacity of the 2D surfels.

Normal Estimation We estimate the normal vectors of our initial 2D Gaussians to align them with the surface they represent correctly. Our method employs a simple PCA-based normal estimation algorithm that relies on a small set of nearest-neighbor points.

Scale and opacity parameters The initial values of the scale and opacity parameters of the Gaussians are crucially impacting the initial reconstruction quality and optimization efficiency. The scale values depend on the distances of the points to their respective neighbors. This ensures that Gaussians are smaller in dense regions and larger in sparser regions. We similarly compute the initial opacity values. The opacities of Gaussians in dense regions are set to higher values than those in sparser regions. This strategy proved very effective at compensating for potential inaccuracies in the generated point cloud. We illustrate the initialization process in Fig 2.

4. Experiments

4.1. Reconstruction of DTU Scans

We conduct our evaluation on the widely used DTU dataset, utilizing the standard subset of 15 scans. To assess geometric fidelity, we employ the Chamfer Distance (mm) as our primary metric. This measure averages the accuracy and completeness of the reconstructed mesh w.r.t. the ground truth, with lower values indicating higher quality.

Table 1 presents the comparative results. The key finding is that our method achieves a mean Chamfer distance of 0.74 mm in just 2 minutes of training.

- **Same Quality, 7× Faster**: When compared to the fully converged vanilla 2DGS (trained for 15 minutes), our method matches its geometric fidelity (0.74 mm vs. 0.80 mm). This demonstrates that the lengthy "densification-and-settling" phase of the standard pipeline is unnecessary if the primitives are correctly initialized. We effectively skip to the final result, achieving a 7× speedup without compromising quality.
- **Superiority at Equal Time**: The impact of our priors is even more evident when comparing methods under the same 2-minute time budget. Without our initialization, vanilla 2DGS fails to converge, resulting in a poor Chamfer distance of 1.48 mm. By contrast, our method is 50% more accurate given the same computational resources, proving that our dense start is the deciding factor for rapid surface extraction.

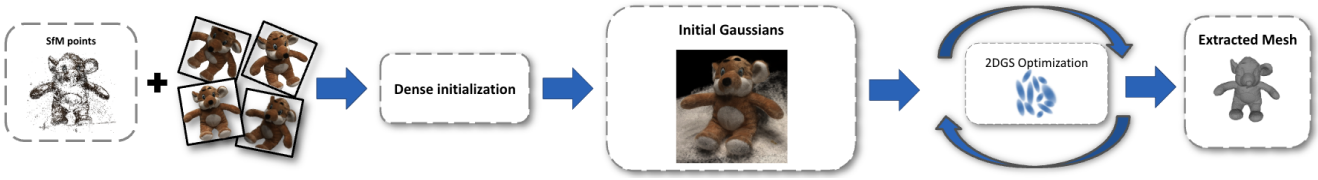


Figure 1. Our method uses SfM points and posed RGB images to generate a dense set of initial Gaussians. We use the iterative 2DGS optimization to further improve the Gaussian reconstruction. The final mesh is extracted through TSDF fusion on rendered depth maps.

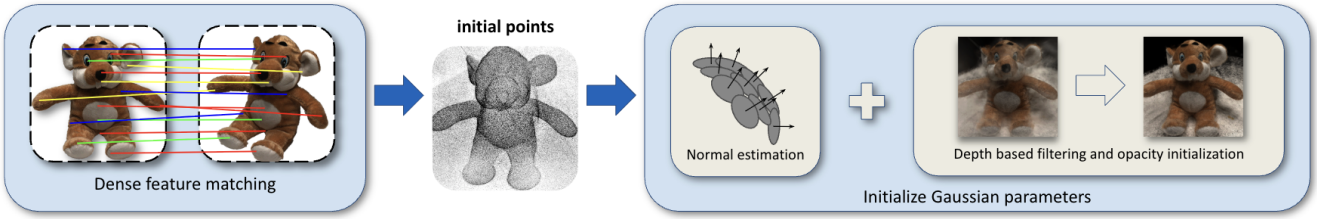
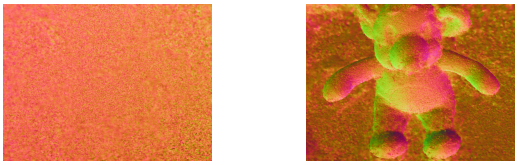


Figure 2. We use dense feature matching to generate a dense point cloud. We then initialize the 2D Gaussians.

4.2. Ablation Studies

4.2.1. Normal estimation

While the dense point cloud from EDGS provides excellent positional coverage, it lacks the orientation information required by 2DGS. Unlike volumetric 3D Gaussians, 2D disks must be correctly oriented to represent a continuous surface. If initialized with random rotations, the disks form a chaotic, disjointed structure rather than a smooth layer, requiring a significantly larger number of iterations to converge. By explicitly initializing 2D Gaussians with estimated normals, we ensure the primitives start in a "surface-like" configuration (see Fig. 3).



(a) Without Normal Estimation (b) With Normal Estimation

Figure 3. Comparison of rendered normal maps of Gaussians after our Initialization with vs. without normal estimation.

Quantitatively, this head-start is significant: after just 2 minutes of training, the model with normal priors achieves a mean Chamfer distance of 0.738 mm, compared to 0.780 mm with random initialization (see Table 2). This performance gap demonstrates that without explicit orientation, the optimizer is forced to expend iterations aligning the disks itself, resulting in significantly slower convergence to a valid surface.

Table 2. Comparison Normal estimation vs. random normals. Chamfer distance (mm) ↓ at 120s.

Experiment	Accuracy	Completeness	Overall
estimated_normals	**0.689**	**0.783**	**0.736**
random_normals	0.748	0.812	0.780

4.2.2. Depth Filter and Opacity Culling

Standard 2DGS initializes primitives with uniform opacity. When applied to our dense point cloud, this treats confident surface points and matching outliers equally, resulting in severe "cloudy" artifacts and floating geometry (see Fig. 4). To mitigate this, we implemented density-aware opacity initialization, assigning high opacity only to points in densely populated regions while suppressing isolated outliers.



(a) Without Depth Filter (b) With Depth Filter

Figure 4. Comparison of rendered initial Gaussians with and without Depth Filter and Opacity initialization.

In addition we utilize a depth consistency check to remove stubborn "floaters" in free space. By aligning relative monocular depth maps with the sparse SfM point cloud, we define a pseudo-ground-truth surface for each view. Points generated by the dense matching that appear significantly closer to the camera than this estimated surface are classified as artifacts and discarded. While opacity tuning provided the most significant visual improvement, the combi-

Table 1. Quantitative results on DTU dataset. We report the Chamfer distance (mm) ↓. We compare fully converged methods (3DGS, SuGaR, 2DGS) with our method after 2 min total runtime. We achieve equal performance in 2 min as 2DGS in 15 min.

Method	24	37	40	55	63	65	69	83	97	105	106	110	114	118	122	Mean	Time
3DGS	2.14	1.53	2.08	1.68	3.49	2.21	1.43	2.07	2.22	1.75	1.79	2.55	1.53	1.52	1.50	1.97	15min
SuGaR	1.47	1.33	1.13	0.61	2.25	1.71	1.15	1.63	1.62	1.07	0.79	2.45	0.98	0.88	0.79	1.32	1h
2DGS	**0.46**	**0.80**	**0.33**	**0.37**	0.95	0.86	0.80	1.25	**1.24**	0.67	0.67	1.24	**0.39**	0.64	**0.47**	0.74	15min
2DGS*	1.38	1.26	1.41	0.93	2.63	2.14	1.25	1.71	1.66	1.30	1.12	2.07	1.22	1.06	1.09	1.48	2min
Ours	0.50	0.89	0.39	0.39	**0.89**	**0.79**	**0.74**	**1.19**	1.36	**0.64**	**0.59**	**1.12**	0.44	**0.61**	0.49	**0.74**	2min

nation of both strategies was necessary to reach our best quantitative result, improving the mean Chamfer distance from 0.770 mm to 0.736 mm (see Table 3).

Table 3. Depth based filtering and opacity initialization. Chamfer distance (mm) ↓ at 120s.

Experiment	Accuracy	Completeness	Overall
with_both	0.701	0.771	0.736
no_depth_filter	**0.690**	**0.762**	**0.726**
no_opacity_tuning	0.760	0.796	0.778
without_both	0.752	0.788	0.770

4.2.3. Normal Regularization

Unlike standard 2DGS, which delays normal regularization, a term to encourage local smoothness and surface continuity, to allow its sparse initialization to organize, we enable these geometric constraints from iteration zero. Because our primitives are pre-aligned via our normal estimation, applying regularization early effectively locks the optimization to the correct surface manifold, preventing geometric drift and accelerating convergence. However, we found this parameter requires careful tuning: while early regularization significantly reduces noise, excessive weight can lead to over-smoothing, washing out fine geometric details.

Table 4. Comparison of different lambda values for normal regularization. Chamfer distance (mm) ↓ at 120s.

Experiment	Accuracy	Completeness	Overall
lambda 0.0	0.737	**0.770**	0.754
lambda 0.01	**0.700**	0.770	**0.735**
lambda 0.05	0.719	0.829	0.774

4.2.4. Densification

A central design choice in EDGS [3] is the elimination of the adaptive densification process (cloning and splitting). While EDGS retains opacity-based pruning, we sought to determine if density control is necessary when optimizing 2DGS for mesh extraction. We evaluated the impact of densification and pruning on the final mesh quality:

- **Densification**: Our experiments revealed that the impact of densification is far less pronounced than that of our geometric priors. However, we observed that densification potentially serves as a valuable safety net for edge cases where the initial stereo matching might fail. In these specific instances, densification helps fill remaining topological voids. Although the average quantitative improvement is minor it is consistent across across the dataset, therefore we retain the mechanism to ensure robustness across all scene types.
- **Pruning**: Complementing densification, we retain opacity-based pruning. We observed that while it does not drastically alter the quantitative metrics on its own, it effectively prevents the accumulation of redundant primitives, maintaining a clean surface representation throughout the optimization.

Table 5. Comparison of reconstruction quality with vs. without densification. Chamfer distance (mm) ↓ at 120s.

Experiment	Accuracy	Completeness	Overall
with_densification	**0.689**	0.783	**0.736**
no_densification	0.702	**0.772**	0.737

5. Conclusion

In this work, we presented an accelerated surface reconstruction framework that leverages dense feature-matching priors to initialize 2D Gaussian Splatting. By replacing the sparse SfM-based initialization with a dense point cloud generated via EDGS-based dense feature matching, and further refining these primitives with normal and depth estimation and density-aware heuristics, we significantly reduce the time required for high-quality mesh reconstruction. Our evaluation on the DTU dataset demonstrates that this informed initialization allows the optimization to achieve competitive surface quality with a speed-up of more than 5x compared to the 2DGS baseline. While our current evaluation focused on object-centric scans, future work could extend this approach to larger, more complex scenes or investigate alternative dense point cloud generation techniques. Overall, our method demonstrates that a dense and geometrically accurate initialization strategy is a critical step toward practical, real-time surface reconstruction.

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